
Rachel's Flowers and decor Shop



This exercise simulates a real-life case where the requirements (the information provided by the client) is not fully complete. This means you must be interactive, discuss the case with your group and ask the instructors for missing information if you need clarification until you come up with the ideal solution.

In this exercise you are a system analyst. Rachel's Flowers and decor owner informed you that they would like to expand their business, to include more items and bouquet on the Shop menu and provide delivery service to their customers.

Your job is to design ER model that caters for the delivery service system for Rachel's Flowers and decor shop and allow business expansion for later.

It is crucial to design a proper system that reflects the below business steps efficiently.

1. Rachel explains to you that the delivery system operation must do the following:
 - When a customer calls and places a delivery order, an employee records the order on a multiform order ticket.
 - The employee captures such details as customer name, business or home address, phone number, order placement time, items ordered, and amount of sale.
 - The multiform document is sent to the bouquet designer who will make the order ready for delivery.
 - multiple copies of the shop's order will be there (one for the client, the shop and one places in a reconciliation box)

- When the order is prepared, the delivery person delivers the order to the customer, removes one order ticket from the delivery box, collects payment for the order, and returns to Rachel's Flowers shop (to take his second order if there are any).
- Upon arriving at Rachel's Flowers and decor shop, the delivery person gives the order ticket and the payment to Rachel.
- Each evening Rachel reconciles the order tickets stored in the reconciliation box with the delivery payments and matching order tickets returned by the delivery person (to avoid any discrepancies).
- At the close of business each evening, Rachel uses the data from the order tickets to update the goods sold and inventory files and generate the daily reports and to order new flowers from her flower supplier (different flower fields) if needed to maintain the stock levels.
- This online system must be able to monitor and report changes inventory material levels and to issue material orders and payments to suppliers. Thus, the central data entity for this system will be an INVENTORY ITEM.
- **Hint:** pay attention to the attributes mentioned above that your ER entities must include (but not limited to, examples customer name, address, phone ..etc)

2. Below is the Data Flow Diagram (DFD) level 0 for the business flow, it is part of the requirements.

Figure 1 Level 0 DFD for Rachel's Flowers and decor Shop



